

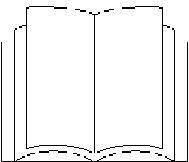


Operator's manual for Attachment



Mini digger

Product number:	400-700-series loaders:	A32393
		A32394
	200-series loaders:	A33153



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The mini diggers are attachment designed for AVANT multi purpose loaders. They are digging attachments meant for occasional digging in places, where the dug soil can be shifted to the sides with the loader. The digging attachment utilises the movements of the loader boom and it is equipped with a hydraulic cylinder to tilt the bucket. The digger can be mounted quickly and it is powerful with respect to its size. Depending on the model, the digger is equipped with either a 250 mm or 400 mm bucket.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Not all maintenance tasks can be performed by the operator; certain tasks must be left to qualified professionals. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the mini diggers

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the mini digger to the loader, ensure that the locking pins of the loader's quick attach plate are in the lower position and that they **have locked the attached to the loader**.
- Start operating with the digger carefully and calmly.
- Keep bystanders away from the operating area of the digger. Do not go near raised digger or under the boom, or allow others to enter the danger area.
- Do not use the digger on horizontally tilted terrains. Ensure the stability of the loader and firmness of the ground also on even terrains.
- Note that the excavation may suddenly cave in. Exercise extreme caution when driving near ditches or embankments, and avoid driving along the ditch, as the machine could suddenly tip over if an edge caves in.
- Pay attention to the contours of the terrain .Pay attention to the surroundings and any other persons and machines moving in the vicinity.



- Always transport the digger as low and as close to the machine as possible to keep the centre of gravity low.
- Shut down the loader engine before any cleaning, maintenance or adjustments.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual.



Note that both the bucket of the attachment and the loader boom move down even if the engine has been stopped. Never go under a raised boom or the attachment. Lower the digger onto ground before leaving the driver's seat.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.

Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.



- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.

4. Mini digger specifications

Listed below are the markings and labels fastened to the mini digger, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.

			
		Ylötie 1 FIN-33470 YLOJÄRVI	
TYPE Axxxxx		No. 00000	
Year 2010		kW	kg

Identification plate of the attachment

- A46771 – Read the instructions before use.
- A46772 – Do not go under a raised attachment; keep a safe distance from raised or freestanding equipment.
- A46803 – Crushing hazard, keep clear of moving parts.



A46771



A46772



A46803

Table 1 – Mini digger technical data

Product number	A33153	A32393	A32394
Working width:	1 200 mm	1 400 mm	
Loading height:	1 500 mm	2 500 mm	
Weight (with bucket):	45 kg	80 kg	90 kg
Standard bucket:	250 mm	250 mm, 2 teeth	400 mm, 2 teeth
Maximum operating pressure:	20 MPa (200 bar)	22,5 MPa (225 bar)	

5. Assembly of the digging attachment

The mini diggers are delivered in working condition. Mount the attachment and its hydraulic hoses according to the following instructions, put the broom in a running condition and practise its use in a safe area. Read the following instructions for mounting, using and maintenance of the mower before starting the equipment.

These instructions are applicable to mounting the attachment to the boom quick attach plate of AVANT loaders that have standard quick attach locking. If your machine is equipped with a hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

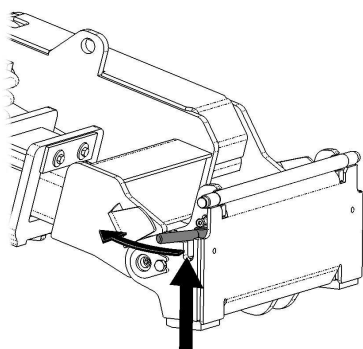
5.1. Mounting the digger to the loader

Mounting the mini digger to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic; the locking latches are released as the locking pins hit the pin guides when the boom front plate is tilted towards the loader (300-700-series) or the locking pin hits the boom (200-series). The attachment is mounted to the loader as follows:

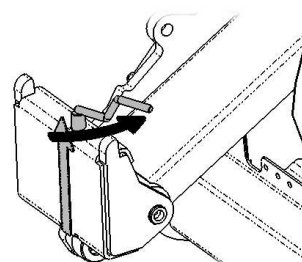


Never move or raise an attachment that has not been locked.

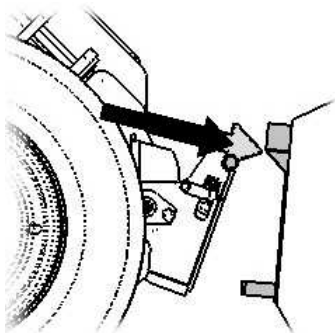


Step 1:

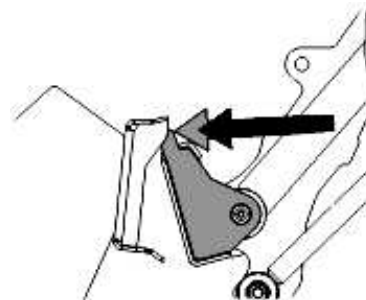
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



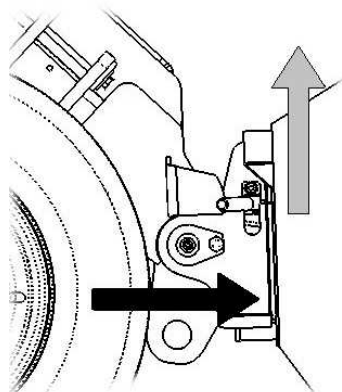
Step 2:



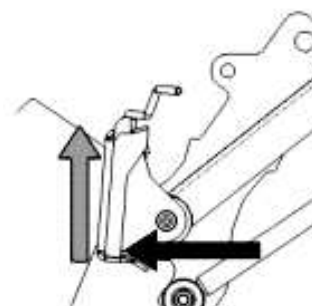
- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.
- Due to the storage position of the mini digger, it may be necessary to assist the mounting of the attachment by hand. Drive the loader of the boom to as close as possible next to the attachment, and tilt the plate fairly to make it easier to lift the digger on the plate.



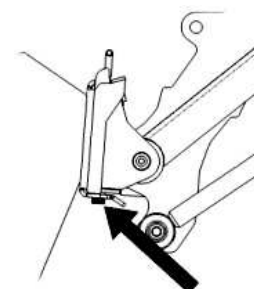
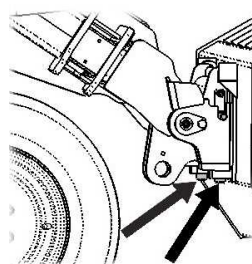
Step 3:



- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Release the locking pins manually.



Before moving or lifting the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may fall on the boom or operator and cause injuries.



Do not use the automatic locking of the locking pins **at a height of over one metre**. If the locking pins do not return to the normal position when the front plate is tilted, do not raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

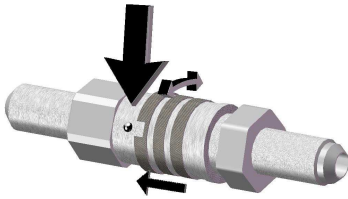
The automatic returning of the locking pins is an useful feature, when attaching an empty attachment at low height and on even surfaces. Otherwise we recommend to lock the pins manually.

5.2. Connecting and disconnecting the hydraulic fittings

The two hydraulic fittings of the mini digger are connected to corresponding fittings on the loader. The hoses should be connected so that the fittings with coloured caps are connected. This way the auxiliary hydraulics control lever will move the bucket the same way every time the digger is used.

To connect and disconnect the fittings, move the collar at the end of the female fitting. Before connecting or disconnecting hydraulic fittings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and the fittings are easy to connect.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause injuries.

When finished working with the digger and storing it, open the bucket completely. This way the digger is easier to attach and remove. Move the auxiliary hydraulics control lever to release residual pressure.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

6. Instructions for use

After the digger and its hydraulic hoses are connected according to the instructions, work can be started. Practise the use of the digger at safe area.

At the first use, the hydraulic hoses and the cylinder may have air trapped in them, which causes the bucket to move inaccurately. The hydraulic system will bleed itself, when the bucket is tilted to its extreme positions a few times.

- Check that all obstacles, including any hidden ones, have been removed from the working area before operation. Remember correct working methods and avoid unnecessary descending from the loader.
- Make sure that it safe to dig in the location. Find out if there are electric cables, water lines or similar at a depth that can be reached with the digger.
- Ensure that the loader and attachment are used in a safe manner and as intended. Do not allow children to operate the machine.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.
- Operate the attachment only from the driver's seat.

6.1. Operation



Check the operating area before using the digger:

- Check that bystanders are at safe distance from the equipment. Do not let others to stay closer than five meters from the equipment, the boom danger area or directly in front of the loader.
- Make sure that there are no electric cables, water lines or similar obstacles within the reach of the equipment.



Beware of the caving in hazard in the excavation or near it. Do not drive near the edge of the hole or allow children to play near it.

The mini digger is operated with the auxiliary hydraulics control lever of the loader. On 200-series loaders, the control device is a pedal on the left and on other models a lever. If the operation of the bucket seems illogical for you, you can change the connection of the hydraulic hoses.

Digging with the mini digger is the most efficient when the edge of the bucket is kept at an angle that allows the edge to cut the soil the most effectively. The bucket should not be pressed too hard downwards, since this would result in lifting the front tires of the loader, and digging becomes ineffective. Pull the edge of the bucket towards the loader. When the bucket has been filled by pulling, close the bucket, and move the material to the side by turning or driving with the loader.

7. Inspections, maintenance and servicing

Continuous maintenance includes regular cleaning and lubrication of the equipment. The condition of the hydraulic hoses and components should be monitored. Due to crushing hazard caused by the movements of the loader boom and the digger bucket, all maintenance work must be done when the attachment has been lowered on the ground and the residual pressure has been released.



All maintenance procedures must be performed when the digger and the loader boom have been lowered down completely. The boom may lower by itself causing serious injuries.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



7.1. Cleaning and storage of the equipment

Clean the attachment regularly. You may use a pressure washer and a mild detergent. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either. Lubricate the lubrication points on the attachment after washing, shown in section 7.2.

Do not let water to stay in the bucket. For longer storage periods touch up the paint where needed, grease the lubrication points and oil the visible parts of hydraulic cylinder shafts to prevent rust damage. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example.

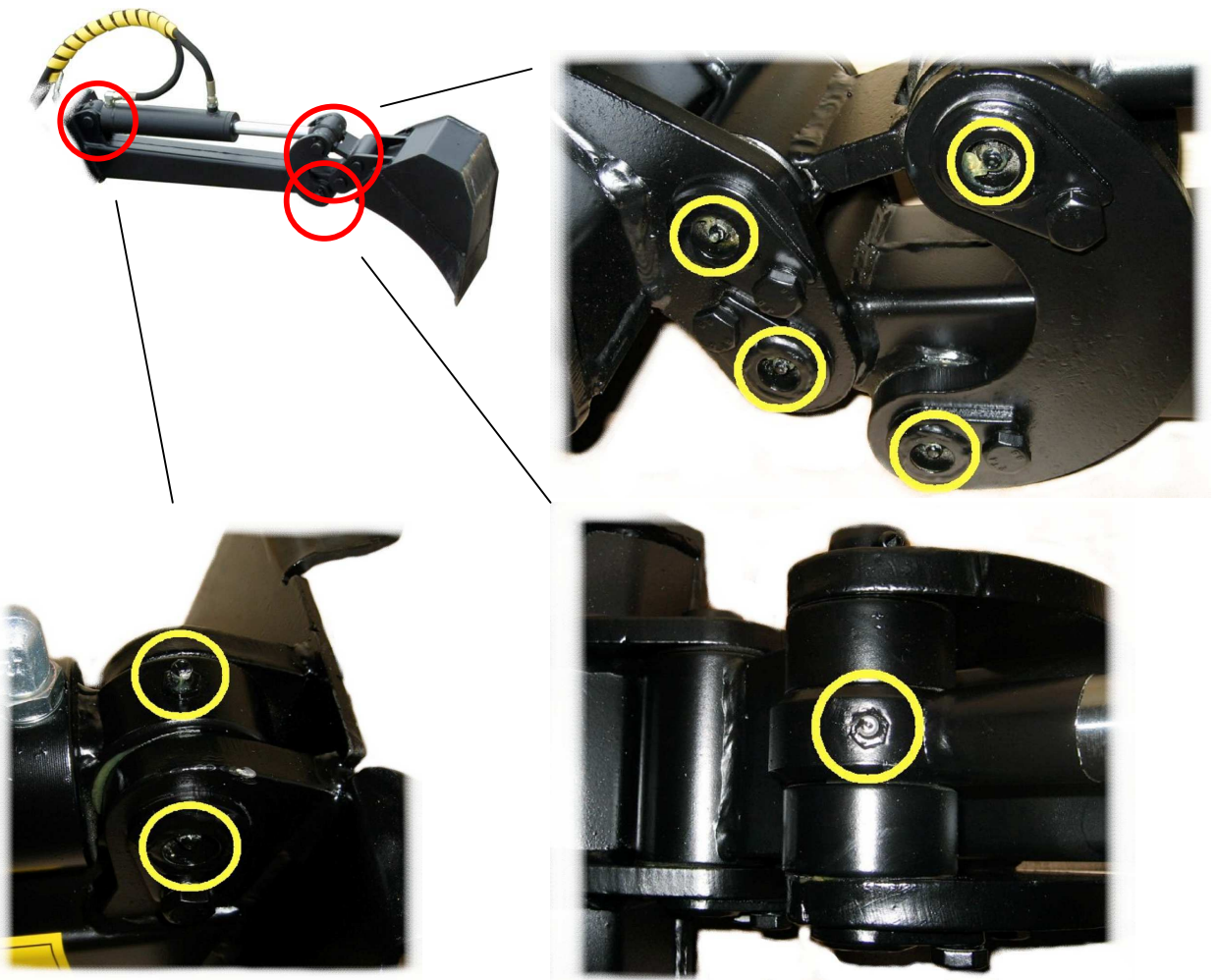
7.2. Lubricating

There are a total of 7 lubrication points on the mini digger for 200-series A33153 and 6 on the models A32394 and A32394. Small amount of lubricant should be added every 5-10 hours of use. All lubrication nipples are standard R1/8" nipples. Replace any damaged nipples.

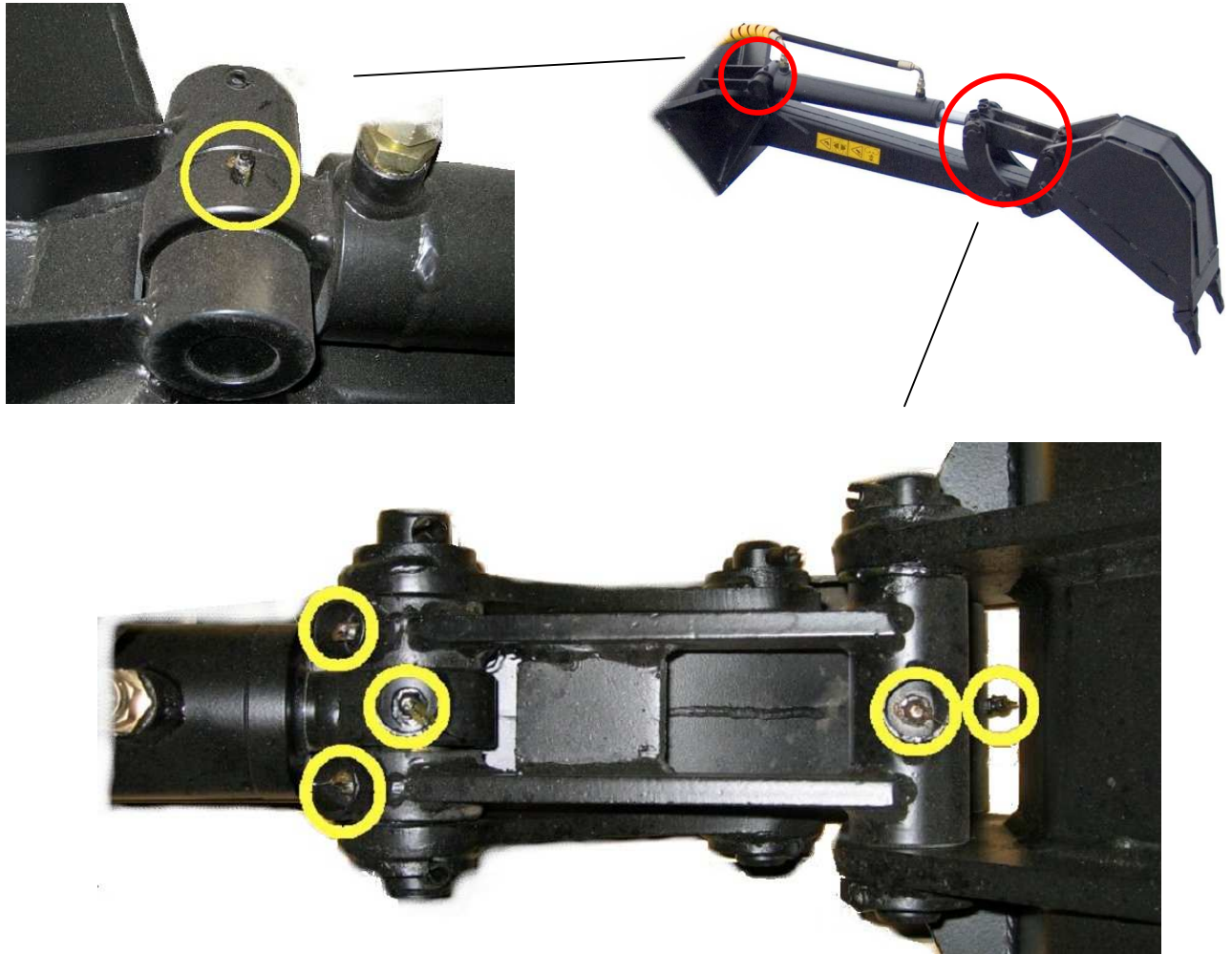


Lubricant must be added approximately every 10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

7.2.1. Lubricating points in the mini digger for 200-series A33153



7.2.2. Lubricating points in the mini diggers for 400-700 -series A32393 A32395



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

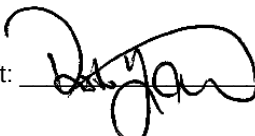
Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Kuvaus: Avant-kuormainten kanssa käytettävä kaivuulaite
 Beskrivning: Grävarm; arbetsredskap för Avant lastare
 Description: Mini digger; attachment for Avant loaders
 Beschreibung: Schaufelarm; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant		200 series	400-700 series
Minikaivuri 120 Grävarm 120 Mini digger 120 Schaufelarm 120	25cm	A33153	-
Minikaivuri 140 Grävarm 140 Mini digger 140 Schaufelarm 140	25cm	-	A32393
	40cm	-	A32394

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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